

An Empirical Evaluation of K-Means-Based Outlier Detection for Zero-Day Intrusion Detection in Industrial IoT Networks

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Abstract: The Industrial Internet of Things environments are getting more vulnerable to attacks that traditional security systems cannot detect. These new attacks are a problem. The current security systems that use learning have a hard time detecting new attacks because they only learn from known attacks. This is an issue in industrial environments where new attacks are happening more often. To fix this problem this research is trying to create a security system that can detect new attacks. This system uses learning and a special kind of detection that can find unusual patterns. The system has a model called Attention-based Bidirectional Gated Recurrent Unit that looks at network traffic to find normal patterns. The system also has a clustering module that groups similar network traffic together. This helps the system find traffic that's unusual and might be a new attack. The system is tested on a dataset of network traffic to see how well it works. The test is done in a way that simulates attacks by not including some types of attacks in the training data. The research looks at how different settings affect the systems ability to detect attacks and avoid false alarms. The goal is to find the settings that balance detecting new attacks with avoiding false alarms. The results show that the settings have an impact, on how well the system works. For example one setting works better for detecting a type of attack while another setting works better for detecting a different type of attack. The Industrial Internet of Things security is a concern because new attacks are happening more often. The current security systems are not good enough to detect these attacks. This research is trying to create a security system that can detect new attacks and protect the Industrial Internet of Things environments. The Industrial Internet of Things security systems need to be improved to detect attacks and avoid false alarms.

Index Terms: Industrial IoT, zero-day detection, intrusion detection, K-Means clustering, outlier detection, ABi-GRU, ToN-IoT.

I. INTRODUCTION

With the proliferation of the Internet of Things (IoT), Industrial IoT (IIoT) has come to the forefront of various critical infrastructures—manufacturing, healthcare, energy, and utilities—where compromise of such networks can lead to disastrous consequences, including physical and financial damage. IIoT networks differ structurally from enterprise IT, which is what most conventional security tools are built for; this has driven rising demand for specialized security measures made for IIoT. Intrusion Detection Systems (IDS) are well established as a means of stopping malicious attacks at their initial phases. However, even with IDSs, network systems are often left open to zero-day attacks that exploit unknown vulnerabilities for which no signature yet exists. This is especially true for IIoT environments, since downtime caused by a missed detection can be extremely costly, in addition to the value of the data itself. Traditional signature-based IDSs are inherently unable to detect zero-day attacks because they can only match known signatures.

Machine-learning-based IDS approaches address this by learning patterns of traffic behavior. Major works have employed supervised learning on labeled attack datasets, which loops back to the original problem: detection is limited to known attack types. Clustering-based anomaly detection was therefore considered, since clusters can indicate unseen attacks without requiring prior labeled examples. Existing ML-based IDS works overcome some limitations of signature-based systems; one framework detects novel attacks via clustering and then updates the classifier, but this update step requires expert labeling, reducing its practicality for IIoT deployments [1]. Another work combines supervised learning with contrastive learning to improve classification and employs KNN and K-Means,

but does not examine how parameter choices affect performance [2]. While such works demonstrate detection accuracy through new models or clustering as an auxiliary feature, little study has been done on the effect of clustering parameters and the balance between false positives and zero-day detection. This motivates a systematic evaluation of centroid-based outlier detection for zero-day intrusion detection in IIoT networks.

To address this gap, this study proposes a hybrid deep-learning-based IDS in which the configuration of K-Means clustering and the centroid-distance threshold are investigated for their effect on the Zero-Day Detection Rate (Z-DR) and false positive rate, evaluated using a leave-one-class-out split on the ToN-IoT network dataset. The study addresses the following research questions:

RQ1: How does the integration of centroid-distance-based outlier detection affect the Zero-Day Detection Rate (Z-DR) of a pre-trained deep learning classifier when identifying unseen malicious flows in a dynamic IIoT environment?

RQ2: How can the Z-DR/false-positive trade-off be balanced based on the distance threshold (d_{min}) and number of clusters (K), and what configuration best balances the two under a leave-one-class-out simulation on ToN-IoT?

The main contributions of this work are as follows:

- A lightweight hybrid IDS that combines an ABi-GRU classification model with centroid-distance-based outlier detection using MiniBatch K-Means for zero-day attack detection in IIoT.
- Under a Leave-One-Class-Out (LOCO) protocol, a systematic assessment of the Z-DR/FPR relationship as a function of

the number of clusters (K) and the centroid-distance threshold (d_{min}).

- Evidence that centroid-distance-based detection performs better for volumetric attacks (DDoS) than for behavioral attacks (Ransomware), and that the silhouette-optimal K does not necessarily give the best zero-day detection performance.

II. RELATED WORKS

Research on zero-day attack detection has produced a range of adaptive, intelligent detection and mitigation techniques aimed at improving cyber resilience in critical systems. This section reviews the literature in four groups: (i) broad AI/ML surveys and empirical benchmarks, (ii) open-set and clustering-based zero-day frameworks, (iii) federated and privacy-preserving detection for distributed industrial settings, and (iv) real-time or domain-specific systems operating outside the standard network-flow case. Dhanushkodi and Thejas [3] combine several techniques, SVM, Random Forest, CNN, RNN, GAN, and Federated Learning, evaluated on traffic generated with CIC-FlowMeter and attacks created with Metasploit, applied across Industry 5.0, IoT, and 5G contexts, reporting 85–100% detection accuracy with SHAP-based explainability. Ozdogan [4] instead empirically tests seven classical classifiers (SVC, Decision Trees, Random Forest, Gaussian Naive Bayes, AdaBoost, XGBoost, and Logistic Regression) on four IoT datasets under different preprocessing and class-balancing strategies, finding that ensemble methods such as XGBoost and Random Forest are most effective, though SVC failed on certain minority classes. Buuri et al. [5] evaluate Microsoft Azure’s AutoML service on two synthetic datasets, obtaining a Voting Ensemble with roughly 95% accuracy. These studies demonstrate the general capability of ML/DL for threat detection but do not target zero-day attacks specifically, nor do they report per-flow processing latency or employ a sequence-aware architecture such as the GRU-based classifier used in this work. Soltani et al. [1] integrate three deep open-set recognition methods (DOC, DOC++, OpenMax) with AutoSVM, clustering, and expert-supervised labeling to address underrepresentation of novel attacks, finding DOC++ performs best under leave-one-attack-out testing on CIC-IDS2017 and CSE-CIC-IDS2018. Touré et al. [6] combine a boosted 1D-CNN for known-attack classification with K-Means clustering and silhouette-based distances for zero-day flows, achieving 96.6–98.4% accuracy on an IBM industrial dataset and NSL-KDD. Duraz et al. [2] propose SECL, combining contrastive learning with regularization (dropout, Von Neumann entropy) to detect and differentiate multiple zero-day attacks, outperforming a fully supervised baseline on the WADI dataset. Sarhan et al. [7] propose a zero-shot learning approach with a Zero-Day Detection Rate (Z-DR) indicator based on Wasserstein distance analysis to explain why specific attack classes go undetected. Across this group, the individual effects of clustering hyperparameters (e.g., k and the distance threshold) on the detection-rate/false-positive trade-off remain poorly explored, and no standardized Z-DR under a LOCO protocol with hyperparameter-level characterization has

been reported. King et al. [8] propose EdgeTorrent, which uses streaming message-passing neural networks and an adversarially trained transformer-based detector instead of hash-based graph-kernel sketches, achieving a perfect F1 score on StreamSpot and throughput exceeding one billion edges per second on an 11-day DARPA dataset. Hadi et al. [9] introduce the GCS-NIDD dataset collected from a real UAV field with three detector tiers (LightGBM, TabNet, Broad Learning System), reaching 93.16% accuracy with low-latency deployment on an NVIDIA Jetson TX2. Yu et al. [10] propose a lightweight GRU-autoencoder combining forward, backward, and patch-based sequence features, scoring anomalies via Mahalanobis distance and achieving over 480,000 packets per second on a Jetson Orin Nano. He et al. [11] map hardware performance counter traces as images to detect zero-days at the processor level, detecting 97% of zero-days from only four hardware events. These four systems collectively show that real-time, resource-constrained zero-day detection is achievable, though each has a distinct application focus: EdgeTorrent [8] operates on host-level provenance graphs rather than network traffic, the framework of Hadi et al. [9] targets UAV communications, the GRU-autoencoder of Yu et al. [10] performs unsupervised anomaly scoring without attack-type classification, and Deep-HMD [11] addresses hardware-based rather than network-based threats. None of these works examine clustering-based outlier detection or report a standardized Zero-Day Detection Rate, motivating the systematic, hyperparameter-level evaluation presented here.

III. PROPOSED METHODOLOGY

The proposed framework evaluates centroid-distance-based outlier detection for zero-day attacks in IIoT using the ToN-IIoT network dataset. Duplicate records are removed and data is categorized, standardized, and split into training and validation sets, then balanced with SMOTE to optimize the K-Means clustering process. A Leave-One-Class-Out (LOCO) protocol simulates the zero-day scenario by withholding the DDoS class entirely from training and validation, using it only for testing. The remaining known-attack classes train an Attention-based Bidirectional Gated Recurrent Unit (ABi-GRU) to classify normal versus known malicious traffic. Concurrently, a K-Means model is trained with the number of clusters (K) chosen via silhouette-score analysis. At inference time, flows whose centroid distance exceeds a threshold d_{min} are flagged as zero-day attacks; the remaining flows are classified by the ABi-GRU. The threshold d_{min} is swept to compare configurations and analyze the trade-off between zero-day detection and false alarms.

A. Dataset Description

The proposed framework was evaluated using the ToN-IIoT `train_test_network` dataset [12], a benchmark IIoT/IoT network intrusion log dataset with 211,043 network flow records across 44 features, including flow duration, packet counts, byte counts, connection state, and protocol-level attributes (DNS, SSL, HTTP). Each flow carries a binary label (0 = benign, 1 = malicious) and a multiclass type label. The dataset contains one normal class (50,000 instances) and nine attack cat-

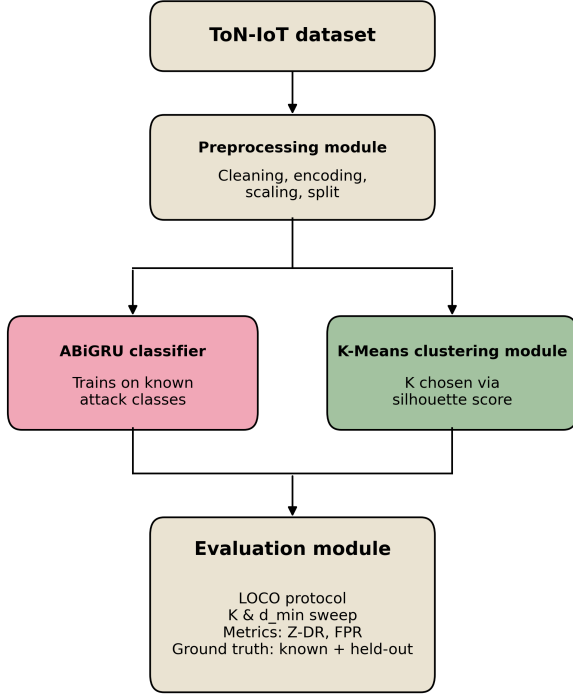


Figure 1. System Architecture of the Proposed ABiGRU and K-Means Zero-Day Detection Framework

egories: backdoor, DoS, DDoS, injection, MITM, password, ransomware, scanning, and XSS, each with 20,000 instances except MITM, which is underrepresented with only 1,043 instances, reflecting realistic IIoT traffic conditions and motivating the resampling strategy adopted during preprocessing.

B. Dataset Preprocessing

Duplicate records and missing data points were removed to preserve data integrity: 20,569 duplicate rows out of 211,043 original records were identified and removed, and no missing values were found. This yielded a cleaned dataset of 190,474 records. Identifier-based features that do not generalize across network session, source/destination IP addresses, port identifiers, and timestamps, were dropped, since these risk overfitting to specific hosts and connections rather than underlying traffic behavior. Categorical features were transformed into binary numerical representations via label encoding, while the `label` and `type` columns were preserved as classification targets.

A Leave-One-Class-Out (LOCO) split was used to emulate a zero-day detection scenario: the DDoS subset was removed entirely from training and validation and reserved only for testing. The remaining samples were split 80%/20% for training/validation using stratified sampling to preserve class proportions. All features were normalized via z-score standardization, with scaling parameters estimated only from the training set to avoid information leakage into the validation and zero-day test sets. The training set was further balanced using the Synthetic Minority Oversampling Technique (SMOTE) to mitigate class

TABLE 1
ToN-IoT `train_test_network` Dataset Class Distribution

Class	Type	# Instances
Normal	Benign	50,000
Backdoor	Attack	20,000
DoS	Attack	20,000
DDoS	Attack	20,000
Injection	Attack	20,000
Password	Attack	20,000
Ransomware	Attack	20,000
Scanning	Attack	20,000
XSS	Attack	20,000
MITM	Attack	1,043
Total		211,043

TABLE 2
Summary of the Preprocessing Pipeline

Step	Description	Resulting Shape
Original dataset	Raw records loaded	211,043 × 44
Duplicate removal	20,569 duplicate rows dropped	190,474 × 44
Missing value handling	No missing values found; no rows dropped	190,474 × 44
Feature normalization	Z-score standardization (fit on training set only)	190,474 × 44

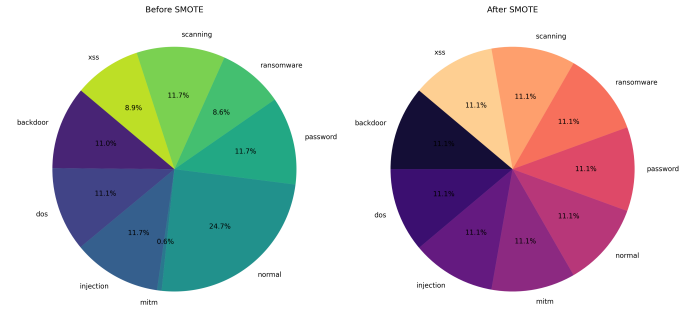


Figure 2. Class Imbalance of Ton-IoT before and after SMOTE was applied

imbalance among known attack categories; the validation and zero-day subsets were left unresampled to preserve their natural distributions for unbiased evaluation of generalization and zero-day performance.

C. Technical Implementation

The K-Means module receives the preprocessed feature vectors and, during training, partitions the feature space into a chosen number of clusters, storing the cluster centroids as representa-

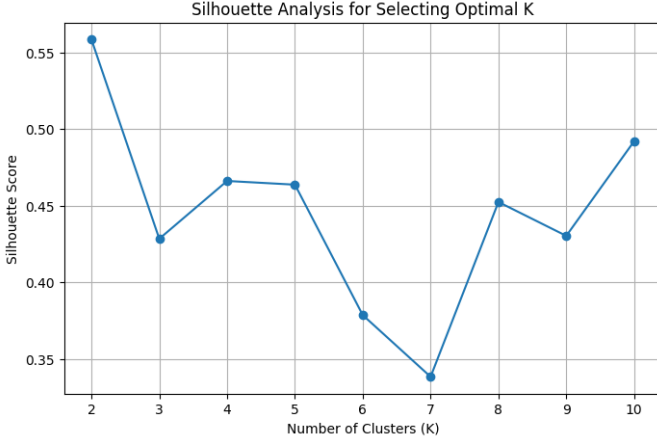


Figure 3. Silhouette analysis for selecting the optimal number of clusters

tive profiles of normal/known traffic. For each network flow, the distance to every centroid is computed, and the minimum such distance serves as an anomaly score reflecting how different the flow is from known traffic. If this minimum distance exceeds the threshold d_{min} , the flow is flagged as a zero-day attack; otherwise, the ABi-GRU classifier assigns it to a known traffic category.

MiniBatchKMeans is used to reduce computational cost, which supports rapid detection in resource-constrained IIoT environments. The centroid-distance anomaly score integrates naturally with the ABi-GRU classifier, which is well suited to sequential network traffic and is computationally lighter than architectures such as LSTMs. The LOCO protocol evaluates the system by withholding one attack type during training, after which the centroid-distance score and the ABi-GRU prediction are combined to reach a decision.

The number of clusters K was swept from 2 to 10, with silhouette analysis (Fig. 3) identifying $K = 7$ as the configuration with the highest silhouette score; this value was used for the main K-Means model. Once trained, the centroid distance of each flow was computed, with the minimum distance to the nearest cluster used as its anomaly score, and a predefined threshold d_{min} used to classify flows as inliers or potential outliers. Applying this centroid-distance approach across the dataset yielded 220,318 flagged outliers and 82,370 non-outliers. The trained MiniBatchKMeans model was evaluated on the validation set and the held-out zero-day set, with the resulting anomaly scores stored for use in the decision-fusion stage alongside the ABi-GRU outputs.

IV. EVALUATION METRICS

Standard classification measures and specialized detection measures were used to evaluate both the known-attack classification and zero-day detection phases. In addition to accuracy, precision, recall, and macro-F1, zero-day detection performance under the LOCO protocol was measured using the Zero-Day Detection Rate (Z-DR) and False Positive Rate (FPR) [7]:

$$Z-DR = \frac{N_{\text{flagged}}^{\text{zeroday}}}{N_{\text{total}}^{\text{zeroday}}} \quad (1)$$

$$FPR = \frac{N_{\text{flagged}}^{\text{benign}}}{N_{\text{total}}^{\text{benign}}} \quad (2)$$

Z-DR indicates the percentage of zero-day (DDoS) flows correctly identified, and FPR indicates the percentage of benign validation flows incorrectly flagged as anomalous. A Combined F1 score treats Z-DR as recall and $(1 - FPR)$ as precision:

$$\text{Combined F1} = 2 \times \frac{Z-DR \times (1 - FPR)}{Z-DR + (1 - FPR)} \quad (3)$$

This metric penalizes configurations that either produce excessive false positives or fail to detect zero-day attacks. The optimum operating point selected for further analysis was the threshold d_{min} that maximized the Combined F1.

V. RESULT ANALYSIS

The ABi-GRU classifier was evaluated for each LOCO split on the validation set of known classes. With DDoS held out (34,097 validation flows), overall accuracy was 94% with a macro-average F1-score of 0.88; with Ransomware held out (34,097 validation flows), accuracy was 93% with a macro-average F1-score of 0.87. Backdoor, DoS, normal, and password traffic were classified with F1-scores above 0.90 in both configurations, consistent with their original class balance before SMOTE oversampling, while MITM was the weakest class in both runs (precision 0.25/recall 0.87/F1 0.39 under the DDoS split; precision 0.24/recall 0.90/F1 0.37 under the Ransomware split).

The confusion matrices for the DDoS-held-out and Ransomware-held-out experiments show the backdoor class classified almost perfectly (5 of 3,742 and 9 of 3,742 misclassified flows, respectively). The greatest confusion occurred between injection and password, and between scanning and XSS, reflecting overlapping feature characteristics: injection produced the most misclassifications (274 cases), XSS contributed 124-138 cases, passwords were frequently misclassified as backdoor (137-139 cases), and scanning showed 130-139 misclassifications followed by backdoor (61-62 cases). MITM consistently showed weak precision (24.7% and 23.0% across the two configurations), with many normal, DoS, injection, and XSS flows incorrectly predicted as MITM-likely because SMOTE-generated samples for the small original MITM class (1,043 instances) overlapped with other classes in feature space, increasing false positives without improving precision.

DDoS results: The overlapping histograms in Fig. 5 confirm that centroid distance alone cannot cleanly separate normal from zero-day traffic at the optimal K , explaining the difficulty of choosing a d_{min} that does not incur a high FPR.

Across $K \in \{2, 3, 5, 7, 9\}$, improving Z-DR consistently increased FPR, with detecting one-third of DDoS flows requiring

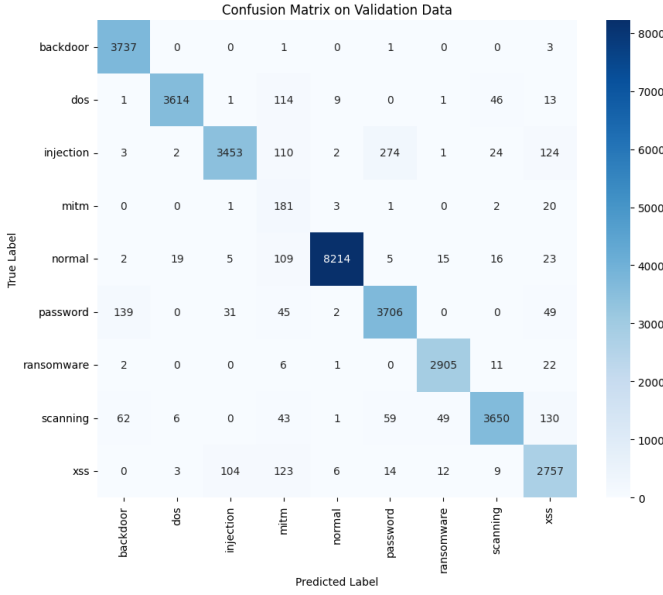


Figure 4. Confusion matrix on Validated Data (DDoS)

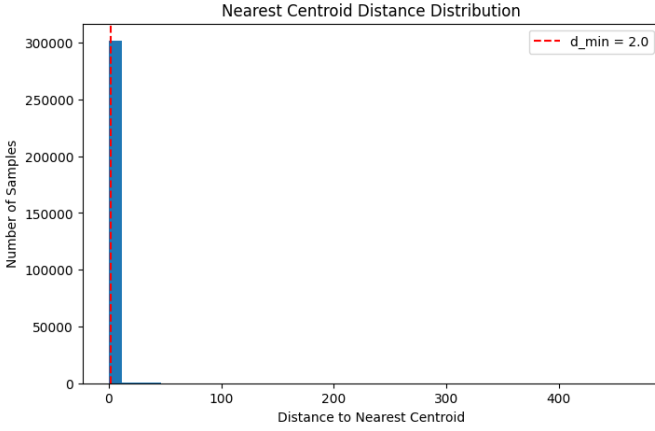


Figure 5. Distance distribution between Normal and Zero-Day flows for DDoS

an FPR above 0.95. Among these configurations, $K = 9$ gave the best trade-off, achieving Z-DR ≈ 0.65 at FPR ≈ 0.90 and Z-DR ≈ 0.24 at the lowest observed FPR (≈ 0.18). This is consistent with the dataset structure, since the nine known classes are better represented by $K = 9$, allowing the withheld DDoS class to remain farther from the learned centroids; lower K values converge to Z-DR ≈ 0.25 – 0.28 , indicating limited separation between known and DDoS traffic. These findings answer RQ2 by demonstrating that the choice of K significantly influences zero-day detection performance.

Ransomware results: In contrast to DDoS, the maximum Z-DR for Ransomware remains below 0.17 until FPR approaches 0.9, with all K values performing similarly except for slight variation near FPR ≈ 0.95 . This demonstrates that centroid-distance detection is considerably less effective for Ransomware than for DDoS, because Ransomware traffic behavior and volume align more closely with normal traffic,

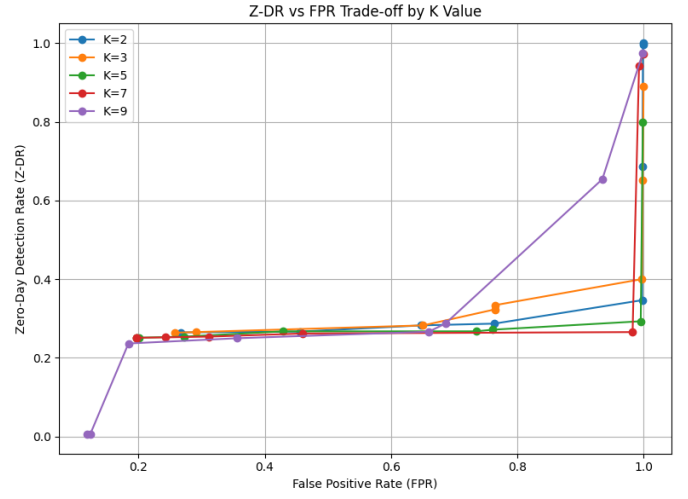


Figure 6. Z-DR (DDoS) versus FPR Trade-off (by K Value)

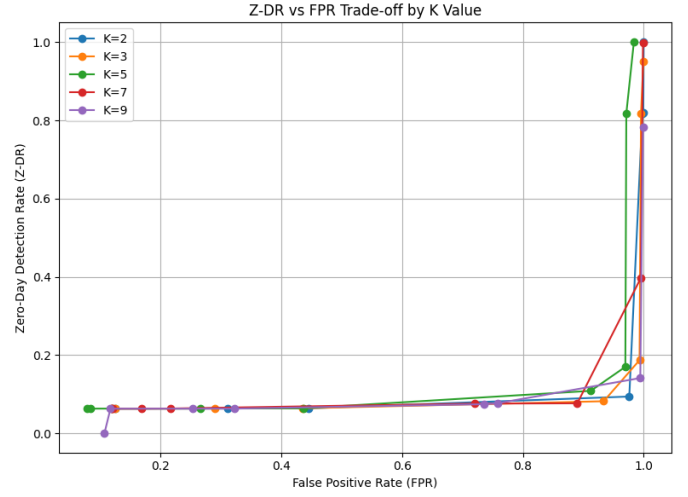


Figure 7. Z-DR (Ransomware) versus FPR Trade-off (by K Value)

whereas DDoS traffic is highly distinguishable due to its volumetric nature. Silhouette analysis selected $K = 2$ for Ransomware, consistent with its weaker Z-DR/FPR profile; Ransomware detection likely requires a different analytical approach than distance-based outlier detection alone.

The Z-DR heatmap (Fig. 8) uses shaded cells to mark configurations with FPR below 0.20, which occur only at higher d_{min} values, where Z-DR is limited to roughly 6–8%. The lowest observed FPR, 0.0786, occurred at $K = 5$, $d_{min} = 5.0$, but with only 6.29% Z-DR. Conversely, at $K = 2$ and $K = 5$ with $d_{min} = 0.5$, a 100% Z-DR was achieved but with an impractical FPR of 0.9838. At $K = 9$, $d_{min} = 5.0$, Z-DR fell to 0.05% with an FPR of 0.1075. As d_{min} increases, FPR decreases steadily while Z-DR stabilizes around 6–7% for most K values above $d_{min} = 3.0$: at $d_{min} = 3.0$, FPR ranges from 0.169 ($K = 7$) to 0.311 ($K = 2$), while at $d_{min} = 5.0$, FPR narrows to between 0.079 and 0.119 for most K values.

K-Means clustering was evaluated using $K \in \{2, 3, 5, 7, 9\}$

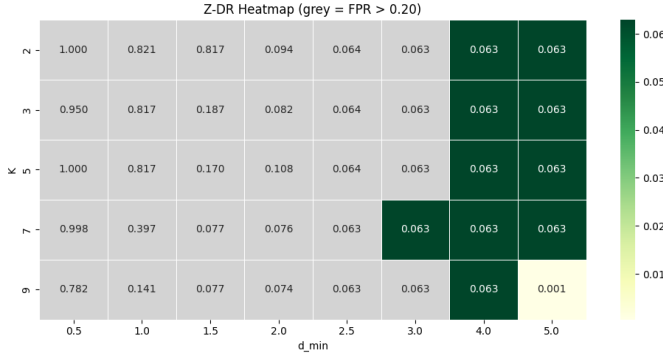


Figure 8. Z-DR (Ransomware) Heatmap for K Values and Centroid-Distance Thresholds.

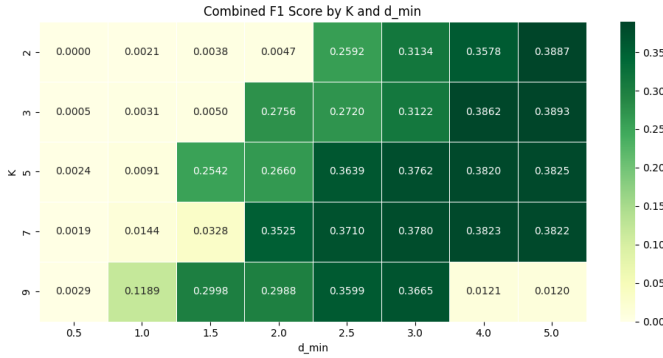


Figure 9. Combined F1-Score by K-value and d_min for DDoS

TABLE 3
Combined F1-Optimal Configuration per Attack Type

Attack (Held Out)	Best K	Best d_{min}	Z-DR	FPR	Combined F1
DDoS	3	5.0	26.37%	25.71%	0.3893
Ransomware	7	2.0	7.57%	72.03%	0.1191

and centroid-distance thresholds $d_{min} \in \{0.5, \dots, 5.0\}$ for both held-out attack types. At low thresholds ($d_{min} \leq 1.0$), both attacks achieved Z-DR above 90%, but with FPR exceeding 0.99, making these configurations impractical. As d_{min} increased, FPR decreased substantially while Z-DR declined at different rates for DDoS and Ransomware. DDoS achieved a markedly better Z-DR/FPR trade-off than Ransomware (Combined F1 of 0.3893 vs. 0.1191): the best DDoS configuration reached a moderate Z-DR ($\approx 26\%$) but still produced an operationally high FPR of about 26%, attributable to DDoS's large volumetric deviations from known-class centroids, whereas Ransomware traffic closely resembles legitimate and known traffic, limiting the effectiveness of distance-based detection. These findings are based on a single DDoS and Ransomware variant within ToN-IoT and may not generalize to other variants. Finally, the silhouette-optimal K did not correspond to the best Combined F1: silhouette analysis selected $K = 7$ for DDoS and $K = 2$ for Ransomware, whereas the highest Combined F1 was obtained

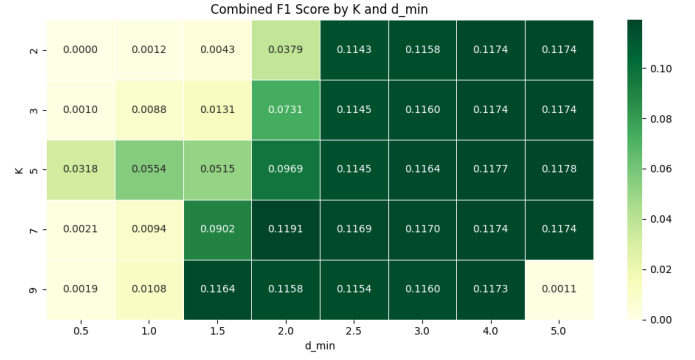


Figure 10. Combined F1-Score by K-value and d_min for Ransomware

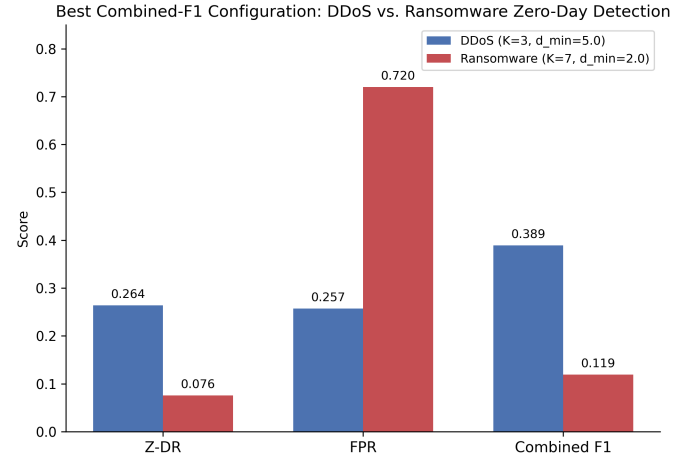


Figure 11. Best Combined-F1 configuration for both attack types (DDoS vs. Ransomware)

at $K = 3$ for DDoS and $K = 7$ for Ransomware.

The reported performance in Table 4 varies because each study uses different datasets, evaluation protocols, and metrics, making direct numerical comparison difficult. Although some models, such as the boosted 1D-CNN + K-Means approach, report higher accuracy on their respective datasets, the proposed ABi-GRU + MiniBatch K-Means framework achieves a competitive 95.27% validation accuracy on ToN-IoT while specifically addressing zero-day intrusion detection. Unlike many existing approaches, the proposed framework combines supervised classification with centroid-distance-based anomaly detection and employs a Leave-One-Class-Out (LOCO) evaluation protocol to assess the Z-DR/FPR trade-off, providing a more realistic evaluation of unseen-attack detection.

VI. DISCUSSION

The pre-trained ABi-GRU classifier showed attack-dependent improvement when combined with centroid-distance outlier detection. Near-total Z-DR at short thresholds offered no real discriminative advantage, since it corresponded to flagging almost all traffic as anomalous. Under a realistic FPR constraint, Z-DR dropped to 26.4% for DDoS and 7.6% for Ransomware: DDoS's extremely high packet/byte volume places

TABLE 4
Comparison with Representative Intrusion Detection Models

Model	Dataset	Reported Performance	Performance
Boosted 1D-CNN + K-Means	IBM Industrial, NSL-KDD	98.4% acc. (IBM), 96.6% acc. (NSL-KDD)	
Deep-HMD	HW performance counter traces	97% zero-day detection rate	
XGBoost (best in Ozdogan)	Four IoT datasets	>97% acc. (DoS)	
PGTAD	IoT multivariate time series	F1 = 0.92	
ABi-GRU + MiniBatch K-Means (proposed)	ToN-IoT	95.27% acc.	validation

it far from known centroids, while Ransomware’s beaconing behavior closely resembles benign traffic, leaving no threshold that achieves both high Z-DR and low FPR for the latter. Centroid-distance detection is therefore effective mainly when the zero-day class is a genuine outlier in feature space. No single (K, d_{min}) configuration was optimal for both attacks ($K = 3$, $d_{min} = 5.0$ for DDoS, $F1 = 0.3893$; $K = 7$, $d_{min} = 2.0$ for Ransomware, $F1 = 0.1191$), because the optimal operating point depends on an unknown attack’s distance from known centroids. Likewise, silhouette-optimal K did not align with $F1$ -optimal K , since silhouette scoring is based only on known-class compactness and does not account for the location of an unseen class. A similar asymmetry has been reported elsewhere, Touré et al. [6] did not perform Z-DR/FPR analysis and reported 96.6-98.4% accuracy, and the Ransomware result here is consistent with Sarhan et al. [7], who found zero-shot detection less useful when the novel class is not clearly separated from training data. Duraz et al. [2] address a similar gap with contrastive learning (SECL) for better class separation on WADI, suggesting that a learned embedding space could help bridge the Ransomware gap observed here. Operationally, DDoS’s best configuration yields approximately 2,500 false alerts per hour per 10,000 flows were high, but potentially manageable. Whereas, Ransomware’s 72.0% FPR would generate over 7,000 false alerts, risking alert fatigue and likely leading operators to disable the detector. The centroid-distance detector is therefore better suited as an auxiliary DDoS signal than as a standalone Ransomware detector. Performance is constrained by three factors: (1) K-Means’ spherical-cluster assumption is poorly suited to skewed distributions; (2) overlapping features between attack categories may require more expressive embeddings; and (3) high dimensionality causes distance concentration. Density-based clustering may help address the first and third issues, while dimensionality reduction (e.g., PCA) may

mitigate the third. These limitations point toward directions for improved representations rather than invalidating the overall approach. Finally, the proposed decision-fusion module was tested with the ABi-GRU and K-Means outputs evaluated independently rather than combined into a single rule; given the low confidence observed on several confused classes, implementing genuine decision fusion is a natural next step, particularly for closing the Ransomware detection gap.

VII. LIMITATIONS

Several limitations remain despite the contributions of this work. First, ToN-IoT is a simulated dataset, so no external validation of the proposed pipeline was performed. Second, only K-Means was used for the clustering layer, leaving other clustering algorithms unexplored. Third, the LOCO protocol withheld only two attack types (DDoS and Ransomware); since this is a simulation, the results cannot fully represent real-world zero-day attacks. Regarding data-related limitations, using SMOTE to correct class imbalance likely shifted centroids closer to the zero-day distribution, directly affecting outlier detection. The high dimensionality of the dataset’s feature space also affects the clustering distribution, as overlapping features for certain attack types can diffuse zero-day prediction ability. Given the modularity of the pipeline, several extensions are possible for future work: a genuine decision-fusion module combining predictions and outlier detection, continual/online learning, integration of multiple IIoT datasets, and industrial deployment and validation.

VIII. CONCLUSION

As IIoT systems expand across critical infrastructures, a persistent security gap remains in detecting unseen cyberattacks. This study contributes: (1) a lightweight hybrid IDS combining an ABi-GRU classifier with centroid-distance-based K-Means outlier detection for zero-day attack identification; (2) a systematic evaluation of K and d_{min} through parameter sweeping; (3) simulated zero-day attack evaluation using a LOCO protocol; and (4) configuration analysis showing the influence of clustering parameters on the Z-DR/FPR balance. This test shows how choosing the values for K and d_{min} really matters when it comes to finding new attacks that have never been seen before. It gives us some ideas, on how to set up a simple system to detect these new attacks in places where machines and computers are connected to the internet like factories and other industrial settings that use something called IIoT.

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